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"Periods" of de Bruijn Sequences

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1. INTRODUCTION

DEFINITION. A *de Bruijn sequence of degree n* (or *span n*) is a periodic binary sequence of periodicity 2^n in which each of the 2^n possible subsequences of length n occurs exactly once (in each periodicity).

Well-known facts (see [1]):

1. There are $2^{2^{n-1}-n}$ de Bruijn sequences of degree n.
2. In each periodicity there are 2^{n-1} 1's and 2^{n-1} 0's.
3. The distribution of runs is the *same* for all de Bruijn sequences of span n.

In each periodicity of length 2^n, there are:

(a) 2^{n-1} runs, of which 2^{n-2} are runs of 1's and 2^{n-2} are runs of 0's.

(b) One-half the runs of each type have length 1,
one-fourth the runs of each type have length 2,
one-eighth the runs of each type have length 3,
and, in general, there are 2^{n-2-l} runs of 1's, and 2^{n-2-l} runs of 0's, for each length l,
$1 \leq l \leq n-2$. (Note that for $l = n-2$, there is *one* run of each type.)

(c) For $l > n-2$, the only runs of length l are *one* run of n 1's and *one* run of n 0's.

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Note that for de Bruijn sequences the *exact* number of runs of length l, for each l, $1 \le l \le n-2$, is the *expected* number of such runs in a *random* binary sequence with $p = q = \frac{1}{2}$ (see [2]).

2. Results on the "Periods" of de Bruijn Sequences of Order $n > 1$

THEOREM 1. *Every de Bruijn sequence of span $n > 1$ has*

$$M_1(\{\text{"periods"}\}) = 4.$$

Proof. In each periodicity of length 2^n, there are 2^{n-2} runs of 1's and 2^{n-2} runs of 0's, hence 2^{n-2} "periods" (a run of 1's followed by a run of 0's). Thus the average length per period is $2^n/2^{n-2} = 4 = M_1$. □

Note. This is the *only* "uniform" regularity. For $n \ge 3$, the moments M_r, for $r > 1$, are *different* among de Bruijn sequences of the same degree. For the moments of the *frequencies*, even the M_1's are *different* among de Bruijn sequences of the same degree $n \ge 3$. (The *frequencies* are the reciprocals of the *periods*.)

THEOREM 2. *In each de Bruijn sequence of degree $n \ge 4$, and for each p with $2 \le p \le n-2$, "periods" of length p occur exactly $(p-1)2^{n-2-p}$ times in each periodicity of the sequence.*

Proof. There are $p - 1$ ways a "period" of length p can occur:

$$\underbrace{\begin{array}{ccc} 1000 & \ldots & 000 \\ 1100 & \ldots & 000 \\ 1110 & \ldots & 000 \\ \ldots & \ldots & \ldots \\ 1111 & \ldots & 110 \end{array}}_{\text{length } p}$$

(The minimum length is 2, since **10** is the shortest possible "period.") For $2 \le p \le n-2$, we can embed the "period" in the pattern

$$\left[011\ldots\underset{\longleftarrow\ p\ \longrightarrow}{10\ldots01}x_1x_2\ldots x_r\right]$$

of length n, with $r = n - p - 2$, and by the "span-n property," there are $2^r = 2^{n-p-2}$ such patterns in each periodicity of the sequence. □

THEOREM 3. *Of the 2^{n-2} "periods" of a de Bruijn sequence of periodicity 2^n, $n \geq 4$, all but $n-1$ of the "periods" have length $\leq n-2$ and are counted in Theorem 2.*

Proof. By Theorem 2, the total number of periods of length p, $2 \leq p \leq n-2$, is

$$\sum_{p=2}^{n-2} (p-1)2^{n-2-p} = 2^{n-2} \sum_{p=2}^{n-2} \frac{p-1}{2^p} = 2^{n-2}\left(1 - \frac{n-1}{2^{n-2}}\right)$$

$$= 2^{n-2} - (n-1). \quad \square$$

Notes. 1. The remaining $n-1$ "periods" have lengths anywhere on the interval $[n-1, 2n]$, and these are different among de Bruijn sequences of the same periodicity.

2. The *sum* of the lengths of these $n-1$ "long periods" is the *same* for all de Bruijn sequences of degree n and equals $n^2 - n + 2$.

3. From Theorems 2 and 3, and Notes 1 and 2 above, fairly tight bounds on the values of M_2 ({"periods"}) can be obtained for the set of de Bruijn sequences of degree n.

3. GENERATING FUNCTION MACHINERY

Define:

$$f(z) = \sum_{n=0}^{m} z^n = \frac{1 - z^{m+1}}{1 - z} \qquad \text{for } |z| < 1. \tag{1}$$

Then:

$$f'(z) = \sum_{n=0}^{m} nz^{n-1} = \frac{-(m+1)(1-z)z^m + (1 - z^{m+1})}{(1-z)^2},$$

$$f'(z) = \frac{1 - z^{m+1}}{(1-z)^2} - \frac{(m+1)z^m}{(1-z)}, \qquad \text{for } |z| < 1, \tag{2}$$

$$f''(z) = \sum_{n=0}^{m} n(n-1)z^{n-2} = \frac{-(m+1)z^m}{(1-z)^2} + \frac{2(1-z^{m+1})}{(1-z)^3}$$

$$- \frac{(m+1)mz^{m-1}}{(1-z)} - \frac{(m+1)z^m}{(1-z)^2}$$

$$= \frac{2(1-z^{m+1})}{(1-z)^3} - \frac{2(m+1)z^m}{(1-z)^2} - \frac{m(m+1)z^{m-1}}{(1-z)},$$

$$f''(z) = \frac{2}{(1-z)^3} - \left\{ \frac{2z^{m+1}}{(1-z)^3} + \frac{2(m+1)z^m}{(1-z)^2} - \frac{m(m+1)z^{m-1}}{(1-z)} \right\},$$

for $|z| < 1$, (3)

$$f'''(z) = \sum_{n=0}^{m} n(n-1)(n-2)z^{n-3}$$

$$= \frac{6}{(1-z)^4} - \left\{ \frac{2(m+1)z^m}{(1-z)^3} + \frac{6z^{m+1}}{(1-z)^4} + \frac{2m(m+1)z^{m-1}}{(1-z)^2} \right.$$

$$+ \frac{4(m+1)z^m}{(1-z)^3} + \frac{m(m+1)(m-1)z^{m-2}}{(1-z)}$$

$$\left. + \frac{m(m+1)z^{m-1}}{(1-z)^2} \right\},$$

$$f'''(z) = \frac{6}{(1-z)^4} - \left(\frac{6z^{m+1}}{(1-z)^4} + \frac{6(m+1)z^m}{(1-z)^3} + \frac{3m(m+1)z^{m-1}}{(1-z)^2} \right.$$

$$\left. + \frac{(m^3-m)z^{m-2}}{(1-z)} \right), \quad (4)$$

$$\sum_{n=0}^{m} nz^n = zf'(z) = \frac{z}{(1-z)^2} - \left(\frac{z^{m+2}}{(1-z)^2} + \frac{(m+1)z^{m+1}}{(1-z)} \right),$$

$$\sum_{n=0}^{m} n^2 z^n = z^2 f''(z) + zf'(z)$$

$$= \frac{2z^2}{(1-z)^3} - \left(\frac{2z^{m+3}}{(1-z)^3} + \frac{2(m+1)z^{m+2}}{(1-z)^2} + \frac{m(m+1)z^{m+1}}{(1-z)} \right)$$

$$+ \frac{z}{(1-z)^2} - \left(\frac{z^{m+2}}{(1-z)^2} + \frac{(m+1)z^{m+1}}{(1-z)} \right),$$

$$\sum_{n=0}^{m} n^2 z^n = \frac{z^2+z}{(1-z)^3} - \left(\frac{2z^{m+3}}{(1-z)^3} + \frac{(2m+3)z^{m+2}}{(1-z)^2} + \frac{(m+1)^2 z^{m+1}}{(1-z)} \right), \tag{5}$$

$$\sum_{n=0}^{m} n^3 z^n = z^3 f'''(z) + 3z^2 f''(z) + zf'(z)$$

$$= \frac{6z^3}{(1-z)^4} - \left(\frac{6z^{m+4}}{(1-z)^4} + \frac{6(m+1)z^{m+3}}{(1-z)^3} + \frac{3m(m+1)z^{m+2}}{(1-z)^2} \right.$$

$$\left. + \frac{(m-1)m(m+1)z^{m+1}}{(1-z)} \right) + \frac{6z^2}{(1-z)^3}$$

$$- \left(\frac{6z^{m+3}}{(1-z)^3} + \frac{6(m+1)z^{m+2}}{(1-z)^2} \right.$$

$$\left. + \frac{3m(m+1)z^{m+1}}{(1-z)} \right) + \frac{z}{(1-z)^2} - \left(\frac{z^{m+2}}{(1-z)^2} + \frac{(m+1)z^{m+1}}{(1-z)} \right)$$

$$= \frac{6z^3 + 6z^2(1-z) + z(1-z)^2}{(1-z)^4}$$

$$- \left(\frac{6z^{m+4}}{(1-z)^4} + \frac{6(m+2)z^{m+3}}{(1-z)^3} \right.$$

$$\left. + \frac{(3m^2+9m+7)z^{m+2}}{(1-z)^2} + \frac{(m+1)^3 z^{m+1}}{(1-z)} \right), \tag{6}$$

$$\sum_{n=0}^{m} n^3 z^n = \frac{z^3 + 4z^2 + z}{(1-z)^4} - \left(\frac{6z^{m+4}}{(1-z)^4} + \frac{6(m+2)z^{m+3}}{(1-z)^3} + \frac{(3m^2 + 9m + 7)z^{m+2}}{(1-z)^2} + \frac{(m+1)^3 z^{m+1}}{(1-z)} \right). \quad (7)$$

4. Evaluation of Sums

THEOREM 4. *By Theorem* 3, *every de Bruijn sequence of span* $n \geq 4$ *has* $n-1$ *"long periods," that is, "periods" of length* $> n-2$. *The sum of the lengths of these "long periods" is* $n^2 - n + 2$. (*This is Note* 2 *following Theorem* 3.)

Proof. We will show that the total length of all the "short periods" is $2^n - (n^2 - n + 2)$, from which the assertion follows. Setting $\frac{1}{2} = z$, the total length of all the "short periods" is

$$L_s = 2^{n-2} \sum_{k=1}^{n-2} (k-1)k \cdot z^k = \frac{1}{z^{n-2}} \left\{ \sum_{1}^{n-2} k^2 z^k - \sum_{1}^{n-2} kz^k \right\}$$

$$= 2^{n-2} \left\{ z^2 \sum_{k=0}^{n-2} k(k-1) z^{k-2} \right\}$$

$$= 2^{n-4} \left(\frac{d^2}{dz^2} \sum_{k=0}^{n-2} z^k \right) = 2^{n-4} \frac{d^2}{dz^2} \left\{ \frac{1 - z^{n-1}}{1-z} \right\}$$

$$= 2^{n-4} \frac{d}{dz} \left\{ \frac{-(1-z)(n-1)z^{n-2} + (1 - z^{n-1})}{(1-z)^2} \right\}$$

$$= 2^{n-4} \frac{d}{dz} \left\{ \frac{1 - z^{n-1}}{(1-z)^2} - \frac{(n-1)z^{n-2}}{(1-z)} \right\}$$

$$= 2^{n-4} \left\{ \frac{-(1-z)^2(n-1)z^{n-2} + 2(1 - z^{n-1})(1-z)}{(1-z)^4} - \frac{(n-1)(n-2)(1-z)z^{n-3} + (n-1)z^{n-2}}{(1-z)^2} \right\}$$

$$= 2^{n-4}\left\{\frac{2(1-z^{n-1})}{(1-z)^3} - \frac{2(n-1)z^{n-2}}{(1-z)^2} - \frac{(n-1)(n-2)z^{n-3}}{(1-z)}\right\}$$

$$= 2^{n-4}\left\{\frac{1-1/2^{n-1}}{1/16} - \frac{(n-1)(1/2^{n-2})}{1/8} - \frac{(n-1)(n-2)(1/2^{n-3})}{1/2}\right\}$$

$$= 2^{n-4}\left\{\left(16 - \frac{16}{2^{n-1}}\right) - 8(n-1)\cdot\frac{1}{2^{n-2}} - 2(n-1)(n-2)\cdot\frac{1}{2^{n-3}}\right\}$$

$$= 2^{n-4}\left\{\left(2^4 - \frac{1}{2^{n-5}}\right) - \frac{n-1}{2^{n-5}} - \frac{(n-1)(n-2)}{2^{n-4}}\right\}$$

$$= 2^n - 2 - 2(n-1) - (n-1)(n-2) = 2^n - (n^2 - n + 2). \quad \square$$

THEOREM 5. *In the de Bruijn sequences of span n where the run of n* 1*'s is immediately followed by the run of n* 0*'s, the* $n-1$ *"long periods" consist of: one period of length* $2n$ *and* $n-2$ *periods of length* $n-1$.

Proof. Such sequences clearly have one "long period" of length $2n$, leaving $n-2$ "long periods" of total length $(n^2 - n + 2) - 2n = n^2 - 3n + 2 = (n-2)(n-1)$. Since the length of a "long period" is *at least* $n-1$, the remaining $n-2$ "long periods" must each have length *exactly* $n-1$. $\square$

THEOREM 6. *The* largest *value of* M_2, *the second moment of the "periods," among all de Bruijn sequences of span n, occurs for those sequences satisfying the condition of Theorem* 5. *The* smallest *value of* M_2 *occurs when the "long periods" consist of* $n-3$ *periods of length n and two periods of length* $n+1$.

Proof. The "short periods" make the same contribution to M_2 for all de Bruijn sequences of span n. It therefore suffices to consider the contribution of the "long periods" to M_2.

There are $n-1$ "long periods," having lengths of at least $n-1$ and at most $2n$, with a total length of $n^2 - n + 2$. Let the "long periods" be $n-1 \le l_1 \le l_2 \le l_3 \le \cdots \le l_{n-1} \le 2n$, subject to $\sum_{i=1}^{n-1} l_i = n^2 - n + 2$, and their contribution to M_2 is $m_2 = \sum_{i=1}^{n-1} l_i^2$. We observe that if $l_i \le l_j$, then $(l_i - 1)^2 + (l_j + 1)^2 > l_i^2 + l_j^2$; because $(l_i - 1)^2 + (l_j + 1)^2 - (l_i^2 + l_j^2) = 2(l_j - l_i + 1) \ge 2$.

Thus, to *maximize* m_2, we make *one* of the l_i's as large as possible, at the expense of the others; while to *minimize* m_2, we make all the l_i as nearly equal as possible, to keep the largest l_i as small as possible. □

5. Remaining Problems

1. Among the $2^{2^{n-1}-n}$ de Bruijn sequences of span n, how many have the unique run of n 1's followed by a run of k 0's, for each k, $k = n, n-2, n-3, \ldots, 3, 2, 1$?

2. What are the exact values of M_2 (the second moment of the lengths of the "periods") which occur for de Bruijn sequences of span n, and how many of the sequences have each of these values?

3. When the run of n 1's is followed by the run of n 0's, this uniquely specifies the lengths (and, in fact, the descriptions) of the other $n-2$ "long periods" of the de Bruijn sequence. What information is sufficient to characterize the entire set of "long periods" for the other de Bruijn sequences of span n?

4. What can be said about the higher moments of the "periods"?

5. What can be said about the distribution of the "frequencies" (i.e., the reciprocals of the "periods") for the de Bruijn sequences of span n?

References

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2. S. W. Golomb, Distribution of periods and frequencies of runs in random binary sequences, *Adv. Appl. Math.* **10** (1989), 278–284.